Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	16 February 2026
Project Name	**Visualization Tool for Electric Vehicle Charge and Range Analysis**
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution tailored to your Tableau EV Dashboard.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	**Data Ingestion & Formatting**	**Import global EV specifications, Import Indian charging station data, Assign geographic roles (Lat/Long)**
FR-2	**Spatial & Range Analytics**	Map plotting of charging stations, Dynamic driving range radius/buffer calculation
FR-3	**Market Performance Visualizations**	Price range comparison chart, Brand/Model distinct count Treemap, Body style distribution bubble chart
FR-4	**Story & Dashboard Assembly**	Compile charts into a unified dashboard, Create sequential Tableau Story points for presentation

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	**The dashboard must feature an intuitive, easy-to-read user interface. Visual cues, tooltips, and clear filter labels must be present so that everyday EV drivers and buyers can navigate the data without technical training.**
NFR-2	**Security**	**The underlying datasets and calculated fields must be locked and protected from unauthorized edits. Only assigned administrators should have the ability to alter the Tableau workbook structure or update the source data.**
NFR-3	**Reliability**	**Spatial calculations (like the distance buffers) and aggregated metrics (like Wh/Km) must consistently output accurate, repeatable results corresponding to the verified source data.**
NFR-4	**Performance**	**The published Tableau dashboard should render quickly. Interactive elements, such as applying a**

		connector type filter or adjusting the range parameter, should update the visual maps and charts within 2-3 seconds to prevent user frustration.
NFR-5	**Availability**	**The final dashboard, once published to Tableau Public or Tableau Server, needs to be accessible to stakeholders and end-users 24/7 across various devices (web browsers and mobile).**
NFR-6	**Scalability**	The data architecture must be flexible enough to easily absorb larger datasets in the future. It should scale to accommodate new global EV brands, additional regions, or live API integration for real-time charging station status without requiring a complete redesign.